Verification of whether the final matrix is in echelon form

1. There are no zero rows in the matrix. $\therefore$ The condition that all the non-zero rows must be above the zero rows is vacuously True.

2. Row 1 Leading element $= 1$ in Column 1
Row 2 Leading element $= 3$ in Column 2
Row 3 Leading element $= -1$ in Column 3
The condition that the leading element of each row is in a column to the right of the columns of the leading elements of all rows above it is True.

3. Row 2 Column 1 $=$ Row 3 Column 1 $= 0$
Row 3 Column 2 $= 0$
The condition that for each pivot column, the elements below the pivot position must be all 0 is True.

$\therefore$ The matrix $\begin{bmatrix} 1 & 0 & -4 & 4 \\ 0 & 3 & -2 & 1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$ is in echelon form.

There are no rows in the matrix of the form $[0\ 0 \ldots\ldots 0\ b]$ where $b \neq 0$.
By Theorem 2, the system of equations is consistent, i.e. there exists a solution set.

$\boxed{\therefore b \text{ is in } W}$ $\quad W = \text{Span}\{a_1, a_2, a_3\}$. If vector b is in W, by defn, b is a linear combination of a_1, a_2, a_3 i.e. $x_1a_1 + x_2a_2 + x_3a_3 = b$ for some $x_1, x_2, x_3 \in \mathbb{R}$. ~~Each of x_1, x_2, x_3 can take infinite~~

c) Show that a_1 is in W.

Soln: $a_1 = 1 \cdot a_1 + 0 \cdot a_2 + 0 \cdot a_3$. 1 and 0 are real numbers. $\therefore a_1$ is a linear combination of a_1, a_2 and a_3. By defn, Span$\{a_1, a_2, a_3\}$ contains the vector a_1. $\therefore a_1$ is in W $\square$